Homework #3 - Problems Key

1a) $5 + 10 + 15 = 30$

%x ⇒ $\boxed{1e}$ $\boxed{0x1e}$ acceptable

1b) $5 - 10 - 15 = -20$

signed int ⇒ 32 bits ⇒ 8 bytes

20 = 010100

-20 ⇒ 101011

101100 ⇒ leading 5 ⇒ 1110 1100

⇓ ⇓

e c

Six more leading f: $\boxed{0xffffffec}$

2) %d - signed int

%ld - long int

%f - float

%lf - double

%u - unsigned int

%lu - long unsigned int

%c - Char

%p - pointer

%s - String

3) fscanf(stdin, "%d %s %f %lf",

&int_input, char_input, &float_input, &double_input);

fprintf(stdout, "%d %s %f %lf",

int_input, char_input, float_input, double_input);

4)

1) 0xe (E is acceptable)

2) 22

3) 0x b10

4) 0x b18

5) 0x b14

6) 0x b20

5)

1) 10 0x 990

2) 0x 14 0x 994

3) 0x 1b 0x 97c

4) 11 0x 994 0x 970

5) 10 0x 990

6) 0x b 0x 994

6)

1) 0x9b0 5 0x9a8

2) 0x9b0 5 0x9a8 0x9e0

3) 0 10 0x950

4) 1 22 0x954

5) 2 17 0x958

6) 3 -5 0x9bc

7 4 31 0x9c0